

Student Security System using RFID and GPS

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Abstract— Throughout our college years, we found campus life to be based extensively on the need to watch our whereabouts and safety. Manual roll-calls or paper logbooks are still used in most schools which are slow, prone to errors and people can cheat on them. Others are more techy solutions such as RFID badge, to make things more automated, but still do not ensure that you are still on campus; you may scan a badge outside the campus and the scanning would still be considered as attendance. Therefore, I have developed an RFID badges and GPS in a single IoT-driven system that proves that it works. I attached an RFID reader and a GPS L80-M39 to an ESP32 DevKit V1. All students receive their own RFID tag and after the tag is scanned, GPS coordinates are obtained at the same time by the ESP32 output and transmitted to a web server running a shipping Node.js and Express. We implemented a few safety measures on the server side: the RFID must not be unknown to the database, the GPS position has to be within prescribed campus ranges (geofencing), we will check against duplicates of scans, and we will establish a time window. It is not until we have done all those checks that we save attendance record. The admin can then access live information on an encrypted dashboard which has role access and clean database. This prototype reduces fraudulent check-in, ensures the campuses are safer, maintains data fresh, and streamlines workflow due to RFID authentication and GPS location verification. My test speed, accuracy, good scaling, and scale-ready modern smart campus are achieved.

Index Terms— Embedded Systems, Attendance Management, Geofencing, Gust Positioning, GPS, IoT, Node.js, RFID, Student Security, Web Application.

I. INTRODUCTION

Keeping an eye on the students and provided campus security are issues that we must cope with on a semester-by-semester basis. The old-fashioned ones, such as rolling call and paper sheets, require alas and Amazon; they are sloppy and occasionally, people get another party to sign them. Students can so easily fiddle in their names even when they are not present. As an increasing number of individuals attend hybrid or online courses, it is becoming more difficult to maintain control of who is actually present in the classroom hence introducing automated and transparent systems is in the offing. Radio Frequency Identification or RFID as it is popularly known has turned into a true game changer when it

comes to tracking individuals in the shortest time possible and not necessarily touching anything. In essence, each student will be assigned a special tag that can be picked up immediately by the reader. Relative to writing the names, RFID saves the time, minimizes errors and it can directly transfer the information directly to a server via the internet The issue lies in the fact that the typical RFID implementation is only aware of a student being who is not necessarily on campus. In a location that requires high security, that blind spot may be dangerous. This is why we are assembling a new system that combines RFID with GPS to ensure that a student is on campus indeed. It consists of a main board which is an ESP32 DevKit V1, an RFID reader and GPS L80-M39. The ESP32 shall be used to read the tag data, retrieve the live whereabouts and transmit all that back to a backend server via Wi-Fi.

There are two checks that the system does, by swiping a student RFID tag, the system will check; the ID against the tag, and the GPS coordinates against a campus geofence, to ensure that they are indeed within campus. Our server was developed using Node.js and Express and stored attendance data in a safe database. To ensure that persons with appropriate rights access or alter the records then we included role-based access control. The web dashboard also allows administrators to monitor live attendance and generate reports as well as make amends as necessary. In addition to keeping a record of those who attend classes, this project ought to ensure that there is a reduction in false attendance in campus as well as enhance responsibility among students. It is designed as a module, and thus can be extended with items such as analytics dashboards, alerts, connections to the school ERP system, and even further down the line, more advanced location checks. We are integrating the magic of embedded hardware and time-tested software development methods to ensure the system is a solid one that can be scalable with the school, and be deployed to any campus a be it a high school, a college or university. This paper explores related workings, describes the architecture, demonstrates how the hardware and software communicate with each other, describes our

assembly process, experiment with it, and discusses future-possibilities

II. LITERATURE SURVEY

The manual systems have been the best-better-beauty when it comes to maintaining the attendance and student security; roll calls, paper registers etc. Even those outdated practices no longer suffice: they can be easily spoiled by proxy attendance, and they will not be able to serve the size of a large university. That is why the movement towards the digital systems in the education management has been strongly shifted towards automated identification techniques.

The very first automated systems were barcodes. They reduce the amount of hand crunching, and you have to point a scanner at them; thus they cannot be skipped by the line of sight, but are simple to copy. Those weaknesses do limit their usefulness to a great degree in large, high-security academic contexts.

The fingerprints, face recognition, etc. Biometric authentication has been researched extensively. Jain et al. [1] put down the fundamentals of biometric recognition and its scalability. The accuracy of biometrics lift authentication but assumes its own hiccups: expensive to deploy, and privacy hiccups, susceptibility to environmental factors, and requires a lot of processing power. Ethical and regulatory red flags are also raised in the storage of sensitive biometric data.

RFID is one of the cool and affordable contactless identification innovations that have been on the rise. Want [12] and Finkenzeller [2] dissect the RFID functionality and its high power. RFID tracking enhances speed and accuracy in the form of passive reading. A number of researches indicate the use of RFID in maintaining attendance within schools and higher educational institutions [9], [10], but most of their systems are local ones that merely store data on campus rather than off.

The IoT has redefined the playing field when it comes to the automation of attendance. Atzori et al. [3], Gubbi et al. [8] and Miorandi et al. [7] provide broad descriptions of IoT architecture and applications. IoT-driven designs allow embedded systems to have wireless pinger servers processed in real-time. Ray [11] and Al-Fuqaha et.al. [19] focus on protocol, tech and scaling issues. These innovations allowed the attendance systems to remain decentralized, as well as be centrally controllable. microcontrollers based on ESP-like ESP32 are a love affair to the internet of things (IoT) due to the inbuilt Wi-Fi and low power consumption. According to the ESP32 Technical Reference Manual [4], it has been said to be perfect in the secure data capture and transmission within a wireless network. Previous research examined RFID systems that relied on microcontrollers [11], [12], but a lot of them were not connected to the scalable clouds or web infrastructure [13]. Transparency and efficiency are provided by online academic management platforms. Brown et' al. [14], Gupta et al. [15], and Chen and Liu [16] all consider centralized attendance tools by which you can monitor and control databases in real time. Cloud structures expand access and ensure data synchronization, in addition to admins having a variety of purposes.

IoT attendance is highly concerned with security and privacy. The research identifies weaknesses in RFID communication and backend servers [17], [20]. The priorities of Sicari et al. [25] and Perera et al. [18] are on the safety and privacy of IoT ecosystems. Sandhu et al. [6] conceptualize role-based access control (RBAC) in order to organise authentication and increase reliability.

Even the RFID authentication protocols are developed to prevent tampering and illegal entry [24]. Ferrag and Maglaras [29] discuss IoT privacy preserving schemes. New research trends involve location verification. Location platform based on IoT [5], GPS-based positioning [21] and geofencing tools [27] provide a sharpening presence of attendance validation that a student is physically within an authorized area. Real-time monitoring and adding time constraints only increase system integrity [20], [21].

Smart campus projects are based on IoT to provide automation and real-time services. Platforms presented by Li and Wang [22] and Ramesh et al. [28] involve the integration of embedded hardware, the centralized management, and analytics. Others also incorporate blockchain to enhance confidence and openness of distributed systems [30]. Nevertheless, with all the technological advantages, a significant number of RFID-assisted attendance manage systems continue to prioritize automation without taking advantage of holistic security, central web management, accurate geolocation, and scalability [22], [23]. Literature adds that we require a common solution that will bring together:

- a) Authentication of identity with RFID.
- b) Tiny IoT processors
- c) GPS-based geofencing
- d) Backend web infrastructure, which is safe and encrypted.
- e) RBAC
- f) Extensible smart campus integration.

Our RFID/GPS Student Safety and Attendance Management System will seal those loopholes. A top-down web architecture and an embedded IoT hardware can be integrated to provide a high-speed, high-quality, and real-time validation to the contemporary campuses.

III. PROPOSED METHODOLOGY

The present paper suggests an organized plan of designing and implementation of an Online RFID- Based Student Safety and Automatic attendance management system. It is aimed at the accurate capture of attendance, robust storage of student database, central administrative tracking and scalability within institutions of higher learning.

The system architecture is based on the step-by-step modular working process, where the sub-systems have a definite job to do in the chain of attendance and safety. The general series will include student ID through RFID, authorization and validation over the server, central database storage and web-based sign-on administration.

Thus, the RFID-based student security and attendance system I have been working on is based on a scalable clientserver architecture. It connects RFID

identification, GPS location checks, and a MERN stack web application to ensure that all of them co-operate with each other effectively. In the entire attendance-acquisition section, a student is assigned a special RFID tag which contains an ID map directly to their record in our database. The USB reader is located at the doors of the classroom and operates at 13.56MHz and when a person scans the UID it is picked by the reader and directly sent to the web interface. There it will be transitioned to the backend to be validated quickly, and the entire operation will take no time, be error-free, and the person who attempts to fugue the attendance will be fruitless. We also connect ESP32 DevKit V1 with GPS L80-M39 module to make the process fat and to ensure that no one marks something on the go. The ESP32 continues to get the real-time coordinates of the device and transmits them back to the server via Wi-Fi using secure Internet protocol HTTPS. At the time a scan occurs, the backend determines whether the GPS points are inside the predefined campus geofence in which they should be before it marks the attendance. That checking twice makes you who you are and where you are.



Fig. 1: ESP 32



Fig. 2: GPS L80-M39

At the backend, all RFID and GPS data is managed using RFID and GPS-friendly NEATful API with Node.js and Express. Upon landing a UID, we validate the student is present in MongoDB, ignore any duplication session records, impose date and time constraints and check GPS zone. The attendance is then only entered in the log after all these checks are passed. We also have role-based access control in which only the admins or the faculty will be able to peek or modify the data. The React based web-app would provide an administration with an intuitive, secure dashboard to enroll students, tap tagging them, call up reports as well as seeing students show up on attendance list with real-time response. It is cross-platform and it is favorable to expand with any contemporary educational arrangement.

Overall, this design would integrate RFID technology, GPS authentication, and a MERN backend to create a stable, efficient,

and safe attendance solution that certainly will be the flavor of the new millennium campus life.

IV. SYSTEM ARCHITECTURE

The RFID-Based Student Security and Automated Attendance Management System proposed is built on a client-server system that can maintain the attendance process high security and real time. The components I used were a USB-based RFID reader, ESP32 DevKit V1 microcontroller, GPS L80-M39, and MERN stack web application. The architecture can be divided into five functional layers with RFID Data Acquisition, Communication, Backend Processing, Database and Web Application each having its own share of exploits in the process of capturing, validating, storing, and administering the attendance.

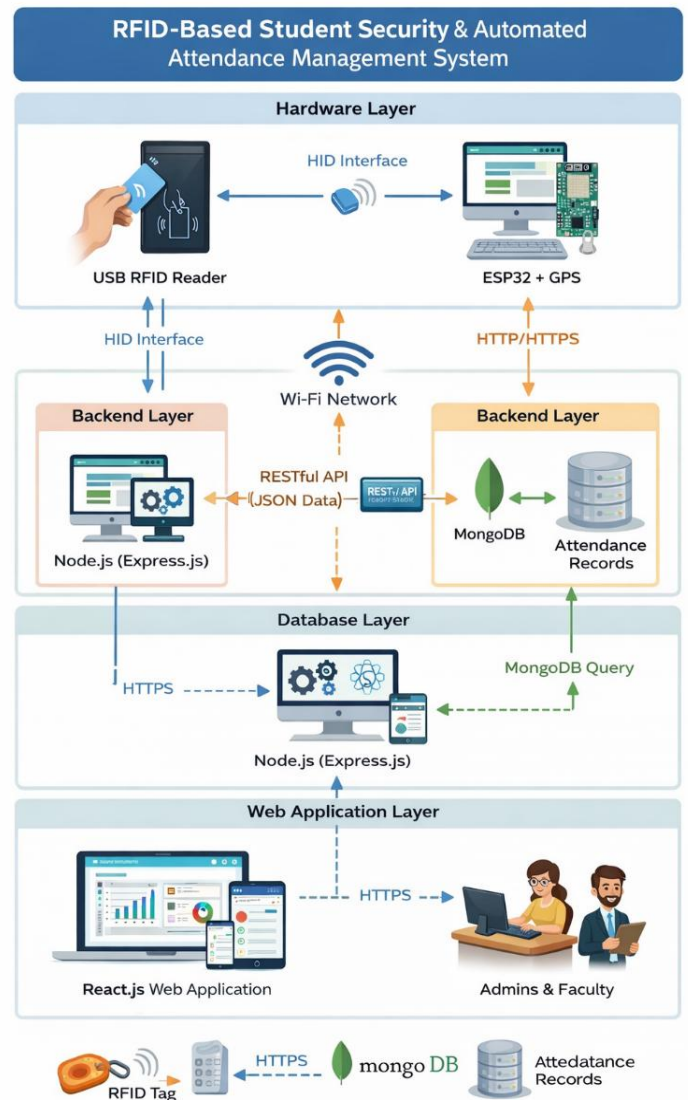


Fig. 3: Overall architecture Student security system using RFID

1. Data Acquisition Layer

In the RFID Data Acquisition Layer, each student will be assigned a different RFID tag that is attributed to the student record. I established a USB RFID reader with frequency set at 13.56MHz at the door of every classroom to capture the UID of the tag as it passes through the reader. Because the reader is running in HID mode it just sends the UID directly to the web-app interface. Meanwhile, an ESP32 together with a GPS L80 M39 continues to retrieve the real-time position of the classroom device thus I can confirm attendance on the basis of location.

2. Communication Layer

The Communication Layer will ensure that the RFID and GPS data are sent to the backend server through HTTP/HTTPS in a secure manner. I structure information in clean JSON and force it using controlled API endpoints. This layer makes the communication fast and verifies requests as valid, and the client devices and the server engage in a safe communication.

3. Backend Processing Layer

The Backend Processing Layer is written in MERN in Node.js and Express. Upon arrival of a UID, a server checks the UID in the MongoDB database of registered students. It contains business logic preventing the occurrence of duplicate entries, authentication of the session time, and checking whether the GPS coordinates are within the specified frame of the geofence. Everything is checked in before the attendance is logged. Role Based Access Control is also hardwired into the system whereby only the authorized users can access the functions of the admins.

4. Database Layer

Database Layer -MongoDB is used to store all student information, RFID mappings, attendance, GPS data, device information, and credentials in a central place. Its document-based architecture makes things scaleable, fasten queries and allows me to make the data flexibly modeled to the needs of multi-classroom structures.

5. Application Layer

The React.js Web Application Layer provides admins a safe but user-friendly interface to manage the student records, monitor live attendance, create reports, modify settings of the system. The web-based UI can be installed on any platform, and the upscale around the school is simple.

All in all, this architecture allows me to automate attendance in a safe and cost-effective way, which involves RFID identification, location checks based on GPS, and a sound MERN backend structure.

V. PROPOSED ALGORITHM

Therefore, this RFID system is simply a means of enrolling real-time attendance. It is not a process of writing everything down, but rather it automates the entire process using RFID tags and a little back-end validation. The connection of the tags to GPS checks and a MERN stack allows it to maintain accurately and reliability and prevents the faking of their attendance by people.

It functions in stages, RFID data is captured, followed by authenticating its data, executing some rule-based checks and storing the data in the database. The UID is captured by the USB reader when scanning the tag and it is sent directly to our web application. The UID is then sent to the backend where we then cross-check it with the student records. This is followed by the system testing a series of conditions - such as whether it is the right time, whether you have been recognized during that session, whether or not the GPS position falls within the university fence with the ESP32 and the L80-M39 GPS sensor.

In case everything is alright, the attendance is registered along with date, time and spot. When it is off, the scan is rejected, this request is recorded to the security team, and an administrator notified via the dashboard. This is how we maintain a safe place and reduce the amount of manual labor, as well as store all the information at one place. Algorithm RFID Student Security and Attendance RFID-Based Student Security and attendance management is provided in algorithm 1.

Input: RFID tag scan

Output: Stored attendance record or scanned attendance record rejected.

1. Start RFID reader and RFID system services.
2. Wait for RFID tag scan.
3. get the distinct RFID UID.
4. Send UID to the backend server through the web.
5. Get relevant student record in MongoDB.

In the event that there is no student record, reject scan and log event.

6. Check the requirements of attendance:
 - a) Check whether scan is made within acceptable time.
 - b) Ask whether the session has been attended or not.
 - c) Check GPS positioning in geofenced area.

In case all the validation requirements are met:

7. Enter attendance with date, time and place of attendance.
8. Revise record of attendance in the database.

If any validation fails:

- a) Deny attendance entry.
- b) Security rejection of logs.

- c) Real-time update of the attendance status in the web dashboard.
- d) Repeat the process of the next RFID scans.

The suggested algorithm can provide safe automation of attendance tallying and preclude duplication and proxy attendance. This is by the integration of RFID authentication, GPS-based location verification and centralized backend validation which improves the reliability and scalability of the system. Moreover, the design as a set of modules will allow introducing future upgrades like analytics dashboards, automatic notifications, and sophisticated monitoring capabilities without making changes to the primary validation process.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

We have experimented with the RFID -based system of student security and attendance. To verify its performance in regards to tracking attendance, response time, location verification, security and scalability, we performed a series of controlled experiments. The test environment consisted of USB RFID readers at the classroom doors, ESP32 DevKit V1 together with GPS L80 M39 module, MERN stack backend and a central MongoDB database.

Student records, RFID tag IDs, and defined set school boundaries were included in the dataset. We conducted various attendance checks at various times and places so as to observe its durationability. Before marking their attendance, each session was to pass both RFID and GPS checks.

It was analyzed that accuracy of the attendance marks was very high when RFID and GPS were simultaneously turned on. The dual validate system would not record attendance unless a valid tag was read within the appropriate zone. Out of zone scans were blocked even with a matching UID into the database. This reduced the rate of bogus attendance and misuse of tags.

TABLE I: COMPARISON OF EXISTING AND PROPOSED SYSTEMS

Metric	ES1 (Manual Method)	ES2 (RFID- Only System)	PS (Proposed RFID + GPS System)
Attendance Accuracy	Low (Human Errors Possible)	Moderate	High
Proxy Attendance Prevention	None	Limited	Strong (Dual Verification)

Processing Time	High	Moderate	Low
Automation Level	None	Partial	Fully Automated
Location Verification	Not Available	Not Available	GPS-Based Geofencing
Duplicate Prevention	Manual Checking	Basic	Automatic Rule-Based
Security Level	Low	Moderate	High
Data Storage	Manual Register	Digital (Basic)	Centralized MongoDB
Scalability	Limited	Moderate	Highly Scalable
Reporting Capability	Manual Reports	Basic Reports	Real-Time Dashboard & Analytics

There was rapid reaction time- attendance requests were processed in real time and the lag between scanning and updating the database was minimal. The IoT device, front-end, and the backend were connected through the API of the rest to maintain the flow of data. The backend procedures such as phone duplicates and time checks had to continue without affect on normal load.

Bad RFID tags were blocked and registered by security checks, which reported them to the admin. In order to maintain clean data in the system, in case one attempted to read the same tag within the same session, the system denied it. Using the central MongoDB to maintain uniform records and to pull attendance whenever it was necessary led to ease of the process.

Scalability investigations demonstrated that the modular structure allows to concurrently operate several classrooms and devices without dropped performance. The addition of more is easily added to a document-based DB and REST services without the need to huge modifications.

Altogether, RFID minting based on GPS geofencing combined with RFID minting proved to be much more reliable and secure in terms of attending than traditional RFID minting or hand-roll. It verifies both your identity and your presence with high-levels of security and efficiency, and offers a solution that can be easily scaled to fit contemporary campuses.

VII. Conclusion

This study includes the design and development of an RFID/ GPS Student Security and Automated Attendance Management System that addresses the problems encountered in the use of traditional ways of taking attendance in schools. The system is a combination of

contactless RFID identification with GPS based location verification, and a MERN based web backend with which we can track the attendance in a safe, precise, and real-time manner without much manual labor.

The system based on the RFID technology to identify students and the GPS geofencing technology to confirm the location of a student can only record the attendance when a student is legitimately in the correct position. Such bi-check prevents the majority of the fake attendance and enhances the campus security. The main backend, which is supported with MongoDB and RESTful APIs, ensures that the data are secure, real time and provides role privilege to administrators. Thus, the entire process is more effective and does not lose the data.

The new system was tested to be processing time and reduces human error by a large margin compared to the old systems such as paper registers, barcode, or standalone biometric systems. The web dashboard helps in controlling, reporting and monitoring the system in real time. The modular system implies that it can be extended and it can serve large and small schools, colleges, and universities.

In conclusion, RFID-GPS attendance is a secure, scalable, and cost-effective solution to use in the contemporary campuses. Further additions may include additional analytics, an app, and extended geofencing, and automatically triggered alerts to make it even smarter throughout the campus network.

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